Kowshik Kesavarapu

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Personal Profile

Results-driven Senior Data Scientist with 7+ years of experience delivering end-to-end ML and Generative AI solutions across enterprise environments. Holds an MSc in Data Science from the University of Surrey. Specializes in Agentic AI, Retrieval-Augmented Generation (RAG), and demand forecasting, with a track record of driving measurable business impact — including $2.4M in annual savings and 85% reductions in manual effort. Experienced in leading cross-functional teams, translating complex business challenges into scalable data solutions, and spearheading GenAI adoption at an organizational level. Proficient in Python, LangChain, LangGraph, and major cloud platforms (AWS, Azure, GCP).

Skills

Programming Languages: Python, SQL

Machine Learning & AI: NumPy, Scikit-learn, PyTorch, TensorFlow, NLTK, Prompt Engineering, RAG , Agentic AI

LLM & AI Frameworks: LangChain, LangGraph, Hugging Face Transformers, LangFuse , Ollama

Web & Application Frameworks: Flask , FastAPI

Cloud & Data Platforms: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Databricks

Data Visualization & BI: Matplotlib, Seaborn, Power BI

Tools & Technologies: Linux, Git, Docker

Methodologies & Soft Skills: Agile/Scrum, Project Management, Stakeholder Management, Team Leadership, Team Management

Work Experience

Innova Solutions - Hyderabad, India

Sr Data Scientist - July 2024 – Current

- Engineered linear programming-based Shipment Recommendation Engine generating optimized truck shipments based on complex constraints, achieving **$2.4M annual savings**, **8% reduction in shipments, and 11% increase in shipped quantities**
- Designed and deployed an AI-driven Excel Template Analyser that automated review of **100+ templates**, improving compliance accuracy from **~65% to 95%** and increasing analyst report delivery capacity from **2–3** reports to **10–15** reports per person — effectively a 5x productivity gain eliminating manual validation entirely
- Architected an Agentic AI platform for the RMG team consolidating 5 legacy systems (SQL, Elasticsearch, REST APIs, SharePoint) into a single conversational interface — reducing candidate identification time from **1 week to near-instant**, eliminating manual lookups across multiple internal tools. Deployed to **200+** users across delivery leadership, senior managers, and project managers, transforming a multi-step manual process into a single natural language query
- Supervised development of Newsletter AI Platform reducing manual newsletter creation time by **85%**, delivering **$10,200** annual savings per group
- Developed multiple RAG chatbots with Natural Language Query (NLQ) capabilities using Hybrid RAG architectures, leveraging LangGraph for agent orchestration and LangFuse for observability, tracing, and performance monitoring
- Spearheaded org-wide GenAI adoption strategy, identifying high-value opportunities and establishing capability-building programs that enhanced productivity across departments
- Developed AFS Stock Prediction Model combining depletion forecasting with stock-out probability classification, enabling proactive inventory management and reducing revenue loss
- Built regression-based ML models using 5 years of historical data for Roll End Date Prediction, improving customer confidence and operational stability
- Architected comprehensive BI infrastructure using Power BI, delivering 30+ custom dashboards with established data governance frameworks

- Led cross-functional team of 12 data scientists and analysts, mentoring interns and coordinating multiple concurrent projects to deliver measurable business results

Cohere | UK

Data Scientist - November 2023 – July 2024

- Led a team of 10 specialists in generating high-quality reasoning and solution datasets for Python and SQL LLM fine-tuning, directly contributing to the training pipeline of Cohere's large language models
- Designed and enforced dataset quality standards and review frameworks, ensuring consistency, accuracy, and reasoning correctness across all submissions before compilation into final training sets
- Architected end-to-end data curation pipelines covering problem design, solution generation, reasoning validation, and final quality control — ensuring training data met the rigorous standards required for effective LLM fine-tuning
- Collaborated with cross-functional teams to translate model capability requirements into concrete dataset specifications, bridging the gap between research needs and practical data generation

Yunex Traffic | UK

Data Scientist - August 2021 – August 2023

- Developed and deployed a Naïve Bayes text classification model to automate ticket categorisation in the internal Incident Management tool, saving 72 hours of manual effort per week across 9 teams and increasing billing frequency from monthly to bi-weekly
- Deployed a license plate detection system on live highway cameras, combining OpenCV for vehicle detection with a Tesseract OCR model fine-tuned specifically for UK number plates — ported and deployed directly onto edge camera hardware for real-time processing
- Built a conversational agent fine-tuned on internal tender data using Llama, integrated with company tools via Hugging Face and Flask — reducing repetitive manual effort in tender generation and improving team workflow efficiency
- Developed a traffic demand forecasting POC using real-time loop data and time series models (Prophet, SARIMA), evaluating forecasting accuracy via MAE, RMSE, and MAPE to establish feasibility for future product development in route optimisation and traffic light timing

Cognizant Technology Solutions | Hyderabad

Analyst - January 2019 – August 2021

- Performed exploratory data analysis (EDA) on large-scale insurance domain datasets, generating actionable reports and insights for stakeholders using Python and SQL
- Engineered Python-based data pipelines to parse and transform legacy COBOL codebases at scale, applying automated pattern recognition and rule-based classification to drive the COBOL 3 → COBOL 4 migration
- Developed and optimised SQL queries and database scripts for large-scale data extraction and analysis on mainframe insurance systems, improving query performance and data reliability
- Identified data quality and system stability issues through structured analysis, implementing permanent fixes that improved overall platform reliability

Education

- MSc Data science from University of Surrey (2021-2023)
- BTech in CSE with specialization of Data Engineering from KL University (2014-2018)